Cambridge Assessment International Education

Cambridge O Level

CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

5034367099

MATHEMATICS (SYLLABUS D) **4024/22**

Paper 2 **May/June 2023**

2 hours 30 minutes

You must answer on the question paper.

You will need: Geometrical instruments

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You should use a calculator where appropriate.
- You may use tracing paper.
- You must show all necessary working clearly.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.
- For π, use either your calculator value or 3.142.

INFORMATION

- The total mark for this paper is 100.
- The number of marks for each question or part question is shown in brackets [ ].

This document has **20** pages. Any blank pages are indicated.

DC (CE/JG) 315946/3
© UCLES 2023

[Turn over

1 **(a)** A shop buys some fruit.
The table shows the bill for this fruit.

Item	Quantity (kg)	Price per kg ($)	Cost price ($)
Bananas	50	0.51	25.50
Oranges	72	1.35	p
Avocados	r	1.95	q
Pears	45	s	51.30
		Total cost price	240.30

(i) Find the value of each of p, q, r and s.

$p =$ .....

$q =$ .....

$r =$ .....

$s =$ ..... [4]

(ii) The shop sells all this fruit for a total of $325.

Calculate the percentage profit.

..... % [2]

© UCLES 2023

(b) In 2022, the shop's total sales were $34 974.

(i) A pie chart is drawn to show the item types that make up these total sales.

(a) The sales for fruit were $9520.70 .

Calculate the angle representing fruit on the pie chart.

..... [2]

(b) The angle representing frozen food is 46°.

Calculate the sales for frozen food.

$ ..... [2]

(ii) The shop's total sales of $34 974 in 2022 were a 4.4% increase on the total sales in 2021.

Calculate the total sales in 2021.

$ ..... [2]

2 **(a)** One chocolate bar costs p cents and one packet of sweets costs 75 cents.
Tanish pays \$9.10 for 5 chocolate bars and 8 packets of sweets.

Form an equation and solve it to find the value of p.
Show your working.

$p =$ ..... [3]

(b) Factorise $6ac - 27c$.

..... [2]

(c) Write $\frac{3m^2n}{10} \times \frac{5n}{9m}$ as a single fraction in its simplest form.

..... [2]

(d) Rearrange the formula $y = \frac{3x^2}{5}$ to make x the subject.

$x =$ ..... [2]

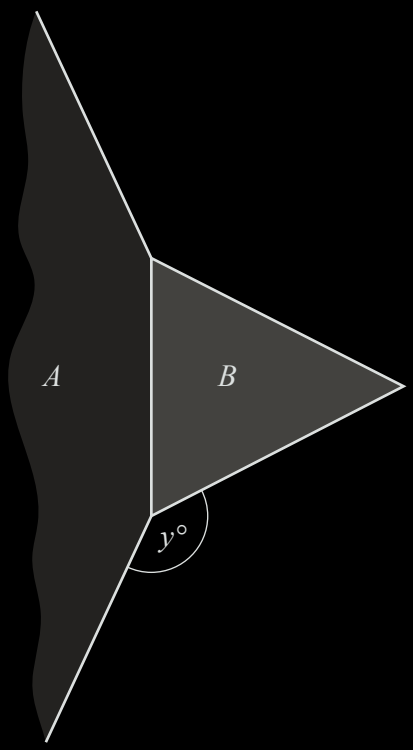
© UCLES 2023

(e) A group of k numbers has a mean of 56.8 .
The number 52 is added to the group.
The new mean is 56.5 .

Find the value of k.

$k =$ ..... [4]

3 (a)



NOT TO SCALE

The diagram shows the equilateral triangle B and part of the regular polygon A which have a common side.
The interior angle of polygon A is 165°.

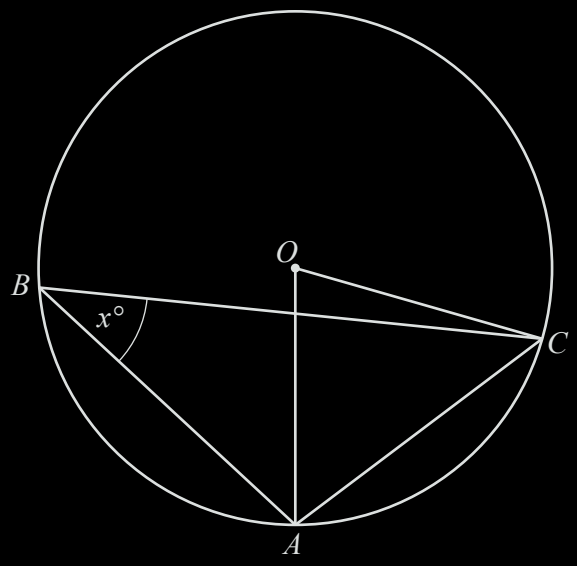
(i) Find the value of y.

$y =$ ..... [1]

(ii) Calculate the number of sides of polygon A.

..... [2]

(b)



NOT TO SCALE

A, B and C are points on the circumference of a circle, centre O.
Angle $ABC = x°$.

(i) Show that angle $OAC = (90 - x)°$.
Give reasons for your answer.

.....

.....

..... [3]

(ii) Angle $BAO = 54°$ and angle $OCB = 11°$.

Find the value of x.

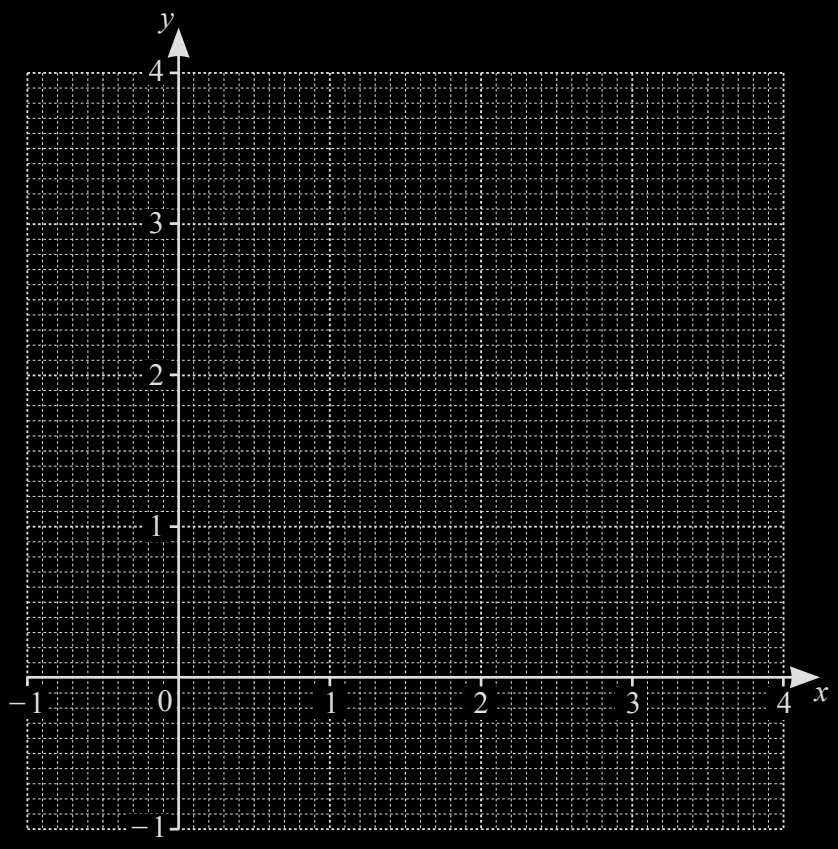
$x =$ ..... [2]

4 **(a)** Complete the table of values for $y = \frac{2^x}{5}$.

x	-1	0	1	2	3	4
y	0.1	0.2	0.4	0.8	1.6	

[1]

(b) Draw the graph of $y = \frac{2^x}{5}$ for $-1 \leqslant x \leqslant 4$.



[3]

(c) By drawing a suitable line on the grid, solve $2^x = 6$.

$x =$ ..... [2]

© UCLES 2023

(d) (i) Complete the table of values for $4y = 2x + 1$.

x	-1	2	4
y			2.25

[1]

(ii) On the grid on page 8, draw the graph of $4y = 2x + 1$ for $-1 \leqslant x \leqslant 4$. [1]

(iii) Find the x-coordinates of the points where the line $4y = 2x + 1$ crosses the graph of $y = \frac{2^x}{5}$.

$x =$ ..... and $x =$ ..... [1]

(iv) The x-coordinates in **part (d)(iii)** are the solutions of the equation $A \times 2^x + Bx + C = 0$, where A, B and C are all integers.

Use the equations $4y = 2x + 1$ and $y = \frac{2^x}{5}$ to find the exact value of each of A, B and C.

$A =$ .....

$B =$ .....

$C =$ ..... [3]

5 **(a)** The table shows the population and area of three countries in 2019.

Country	Population	Area (km^2)
Sri Lanka	2.18×10^7	6.56×10^4
South Korea	5.17×10^7	1.00×10^5
Pakistan	2.17×10^8	8.82×10^5

(i) Write down the value of the smallest population.

..... [1]

(ii) Find the difference in area between Sri Lanka and Pakistan.
Give your answer in standard form.

..... km^2 [1]

(iii) The population density of a country is the number of people per square kilometre.

Find the value of the largest population density from these countries.

..... people/km^2 [2]

(b) In standard form, $A = 8.6\times10^n$ and $B = 1.5\times10^{n-1}$.

Giving your answer in standard form, find in terms of n

(i) $A - B$

..... [1]

(ii) $A \times B$.

..... [2]

© UCLES 2023

6 Sophia takes part in the Trio Challenge.
She walks, then cycles and then swims.

Trio Challenge
Walk 6.3 km
Cycle 3000 m
Swim 1800 m

(a) Write these distances walk : cycle : swim as a ratio in its simplest form.

..... : ..... : ..... [2]

(b) Sophia walks at an average speed of 1.4 m/s.
She completes the walk at 11 05.

Find the time she starts walking.

..... [3]

(c) Sophia cycles a distance of 3000 m correct to the nearest 10 metres.
She cycles this distance in a time of 450 seconds correct to the nearest 10 seconds.

Calculate the upper bound for her average cycling speed in metres per second.

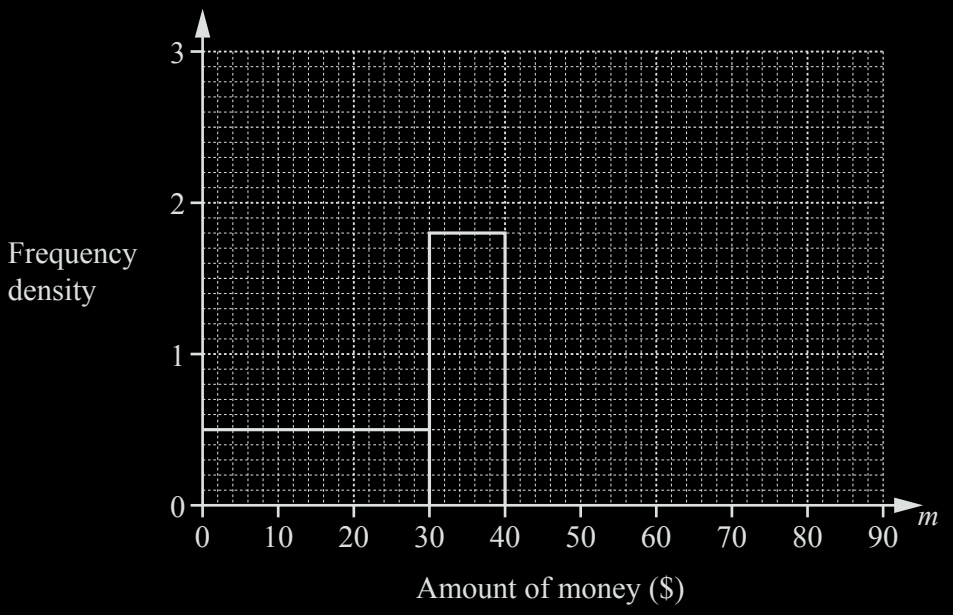
..... m/s [3]

© UCLES 2023

7 **(a)** On Monday, the amount of money spent on a website by each customer was recorded. The table shows the results.

Amount of money ($\$m$)	$0 < m \leqslant 30$	$30 < m \leqslant 40$	$40 < m \leqslant 50$	$50 < m \leqslant 60$	$60 < m \leqslant 90$
Frequency	p	18	24	19	24

The histogram shows some of the results.



(i) Find the value of p.

$p =$ ..... [1]

(ii) Complete the histogram. [3]

(iii) One of these customers is selected at random to receive a discount voucher.

Calculate the probability that this customer spent more than \$50 on Monday.

..... [1]

© UCLES 2023

(b) The table shows the amount of money spent on a website by each customer on Tuesday.

Amount of money ($\$m$)	$0 < m \leqslant 30$	$30 < m \leqslant 40$	$40 < m \leqslant 50$	$50 < m \leqslant 60$	$60 < m \leqslant 90$
Frequency	22	16	24	19	14

(i) Calculate an estimate of the mean.

$ ..... [3]

(ii) An error was made and one of the sales on Tuesday was not included in the table.
That customer spent $41.
Tristan says:

> If that value had been included in the table, then the **estimated** mean would have been higher.

Without calculation, explain why he is correct.

.....

..... [1]

8 (a)



NOT TO SCALE

The length of the rectangle R is twice its width.
Rectangle R has a perimeter of 20.4 cm.

(i) Find the length and width of the rectangle R.

length = ... cm

width = ... cm [2]

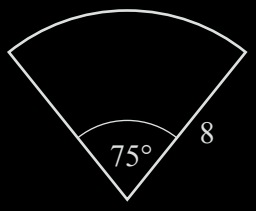
(ii) Rectangle S is mathematically similar to rectangle R.
Rectangle S has a perimeter of 30.6 cm.

Calculate the length of rectangle S.

length = ... cm [2]

© UCLES 2023

(b)



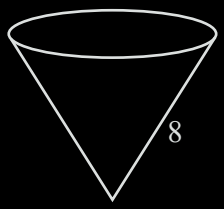
NOT TO SCALE

A piece of card is a sector of a circle with sector angle 75° and radius 8 cm.

(i) Find an expression, in terms of π, for the arc length of the sector.
Give your answer in its simplest form.

..... cm [2]

(ii) **[Volume of a cone $= \frac{1}{3}\pi r^2 h$]**



The piece of card forms the curved surface area of a cone.
The cone is filled to the top with water.

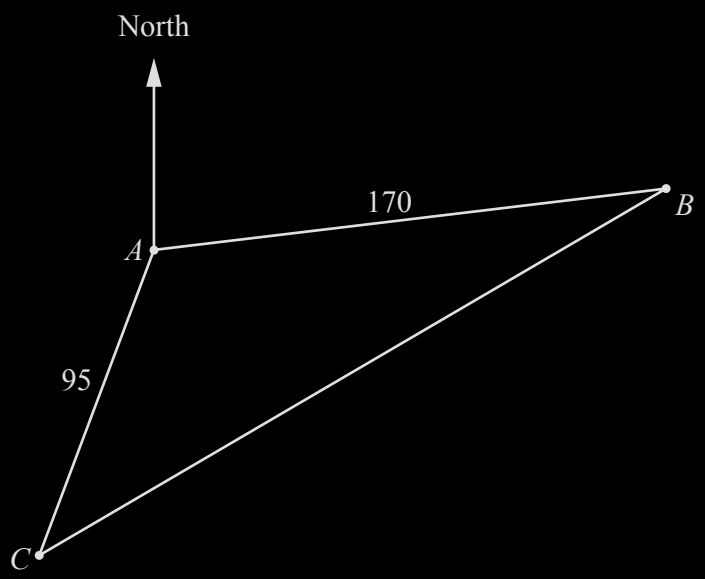
Calculate the volume of water in the cone.

..... cm^3 [5]

© UCLES 2023

[Turn over

9



NOT TO SCALE

A, B and C are points on horizontal ground.
The bearing of B from A is 072°.
The bearing of C from A is 205°.
$AB = 170$ m and $AC = 95$ m.

(a) Calculate BC.

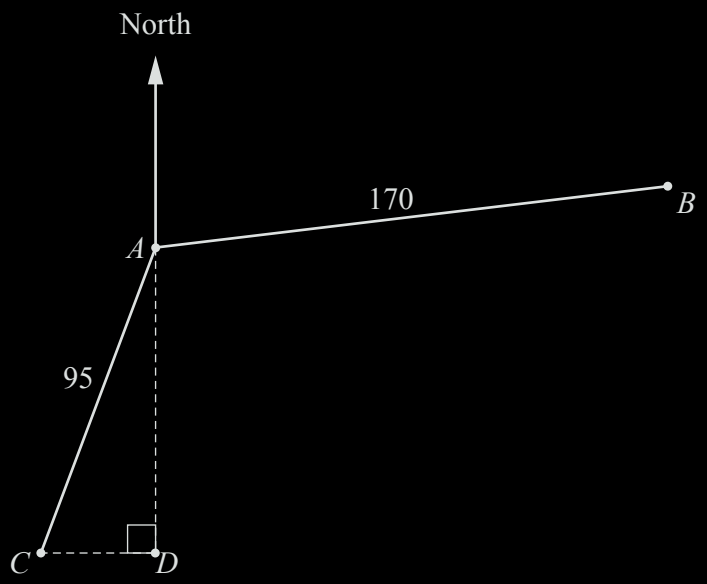
..... m [4]

(b) Find the bearing of A from C.

..... [2]

© UCLES 2023

(c)



NOT TO SCALE

The point D lies on the horizontal ground, due south of A and due east of C.

(i) Show that $AD = 86.1$ m, correct to 1 decimal place.

[2]

(ii) A point X is at the top of a vertical mast at A.
The angle of elevation of X from B is 7°.

Calculate the angle of elevation of X from D.

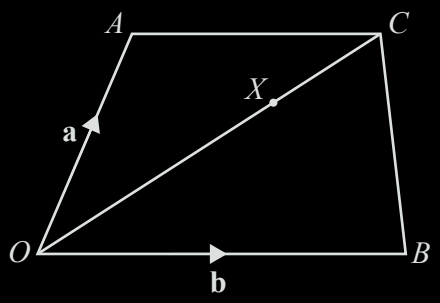
..... [4]

10 **(a)** F is the point $(6, 1)$, G is the point $(-2, 4)$ and $\overrightarrow{GH} = \begin{pmatrix} -1 \\ -8 \end{pmatrix}$.

Calculate $\left|\overrightarrow{FH}\right|$.

$\left|\overrightarrow{FH}\right| =$ ..... [3]

(b)



NOT TO SCALE

$\overrightarrow{OA} = \mathbf{a}$, $\overrightarrow{OB} = \mathbf{b}$ and $\overrightarrow{AC} = k\mathbf{b}$.
X is the point on OC such that $OX = mOC$.

(i) Write $\overrightarrow{OX}$ in terms of m, k, **a** and **b**.

$\overrightarrow{OX} =$ ..... [2]

(ii) $\overrightarrow{BX} = \frac{3}{5}\mathbf{a} - \frac{1}{2}\mathbf{b}$

Find the value of k.

$k =$ ..... [3]

© UCLES 2023

BLANK PAGE

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge Assessment International Education Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cambridgeinternational.org after the live examination series.

Cambridge Assessment International Education is part of Cambridge Assessment. Cambridge Assessment is the brand name of the University of Cambridge Local Examinations Syndicate (UCLES), which is a department of the University of Cambridge.

© UCLES 2023